Guide de Diagnostic : Dommages sur les Contacts Dorés (Gold Pins) de la RAM

1. Contexte et Description Technique

Ce document technique détaille l'inspection visuelle, les causes électriques et les procédures de remise en état des connecteurs dorés (Edge Connector Pins) sur les modules de mémoire vive DDR3, DDR4 et DDR5.

2. Symptômes Observés sur la Machine

Arrêt immédiat de l'alimentation (court-circuit), odeur de brûlé, non-détection de la barrette dans le BIOS, ou redémarrages en boucle (boot loop).

3. Protocole de Diagnostic et Résolution

1. Isoler électriquement la machine et extraire la barrette.

2. Inspecter les 288 broches (DDR4/DDR5) sous loupe pour repérer les traces de carbonisation noire, d'oxydation verte ou de pâte thermique isolante.

3. En cas d'oxydation légère ou de pâte thermique, nettoyer à l'alcool isopropylique (IPA 99%). Si une piste dorée est fondue ou arrachée, le module doit être mis au rebut et le slot carte mère inspecté.

4. Articles et Retours d'Expérience Extraits du Web

Extrait de [Ram Gold Pin Damage Check (memory Stick Inspection)] :
RAM Gold Pin Damage Check (Memory Stick Inspection) If a RAM stick has damaged, dirty, or corroded gold contacts, stop using it and inspect both contact faces with the computer fully powered off and unplugged. Do not scrape or polish the contacts. Test a clean-looking module in a known-good compatible slot, one stick at a time, and replace it if plating is missing or damage is deep. A common mistake is to keep restarting a computer that will not boot, or to rub the contacts with an eraser in hopes of a quick fix. Either move can make the problem worse. First remove power, then inspect the memory stick and work out whether the fault follows the stick, the slot, or a memory setting. Start with safe triage Safe triage means preventing further electrical damage before testing memory. Power down the computer, disconnect AC power, and follow the system’s service instructions before opening it. If liquid reached the computer, do not switch it on to “see if it works.” A wet or contaminated slot can short when powered. If a laptop or desktop has a swollen, hot, hissing, or smoking battery, stop handling the computer. Move away from it and seek emergency help if it is smoking or burning. A b...

Extrait de [Corroded RAM Contacts (Isopropyl Alcohol) - LaptopJudge] :
Corroded RAM Contacts (Isopropyl Alcohol) If your PC shows memory errors after a spill or rough handling, shut it down and unplug it before inspecting the RAM. Test one module at a time at default settings to see whether errors follow a module or slot. Clean only light surface residue with 90% or stronger isopropyl alcohol, then let it dry fully. A PC that will not boot can make a small problem feel like a total loss. Yet a memory error does not prove the RAM contacts are corroded. A loose module, a damaged motherboard slot, or an unstable memory profile can look much the same. I start with isolation, not cleaning. That matters because alcohol can remove some surface residue, but it cannot replace worn plating or repair a cracked slot. The steps below help you protect the computer, narrow down the fault, and decide when a home repair has reached its limit. Triage: protect the PC before inspecting RAM Triage means taking the first steps that reduce the chance of more damage before you diagnose the fault. For suspected liquid exposure, power and time matter: shut the PC down, disconnect power, and do not keep trying to boot it. Drying the outside alone does not confirm that internal...

Sources Web Associées :

• RAM Fingers: Repair Gold plated connectors - YouTubeRam Gold Pin Damage Check (memory Stick Inspection)Corroded RAM Contacts (Isopropyl Alcohol) - LaptopJudgeA High-Stakes Repair: Seeking Advice on the Delicate Task of ...How to Repair RAM Memory [Complete Diagnostic & Fix Guide]How to Repair PCB Gold Fingers: 3 Proven MethodsCorrosion issue on the Ram slot - HP Support Community (https://www.youtube.com/watch?v=hlJ-B-wK46k)

- Ram Gold Pin Damage Check (memory Stick Inspection) (https://laptopjudge.com/ram-gold-pin-damage-check/)
- Corroded RAM Contacts (Isopropyl Alcohol) - LaptopJudge (https://laptopjudge.com/corroded-ram-contacts/)

Catalogue Visuel : 10 Photos Réelles (01_ram_gold_pins_damage)

Ces 10 cas visuels issus du dataset illustrent les motifs caractéristiques de cette classe d'erreur :

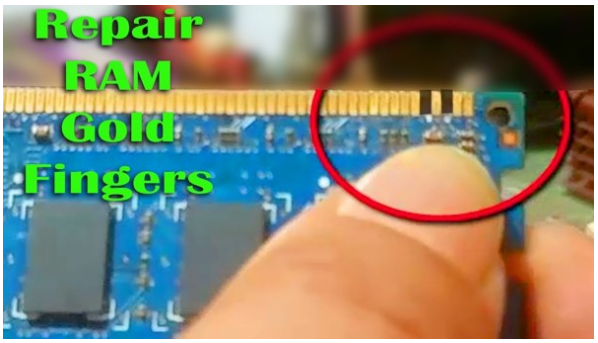


Figure 01 — 01_ram_gold_pins_damage (Échantillon #1)

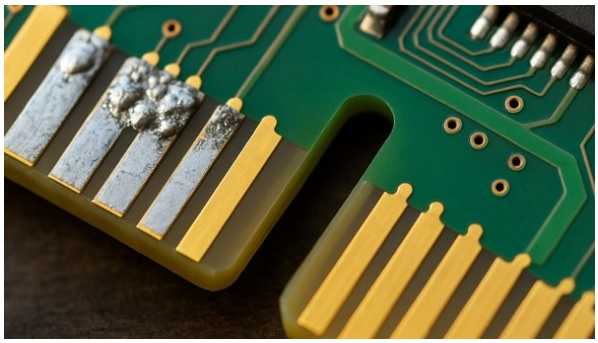


Figure 02 — 01_ram_gold_pins_damage (Échantillon #2)



Figure 03 — 01_ram_gold_pins_damage (Échantillon #3)

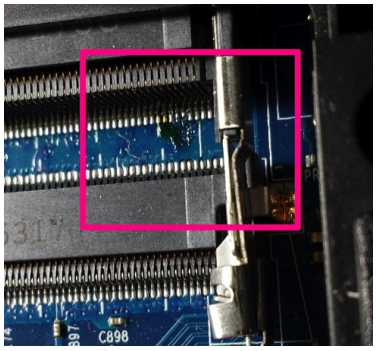


Figure 04 — 01_ram_gold_pins_damage (Échantillon #4)



Figure 05 — 01_ram_gold_pins_damage (Échantillon #5)

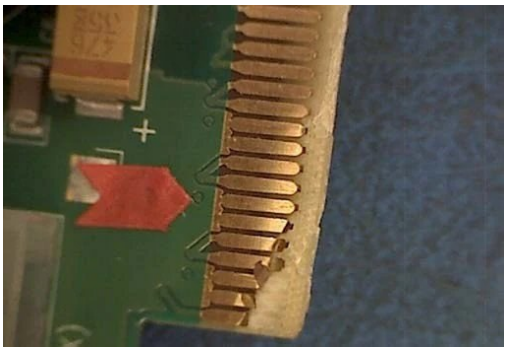


Figure 06 — 01_ram_gold_pins_damage (Échantillon #6)



Figure 07 — 01_ram_gold_pins_damage (Échantillon #7)

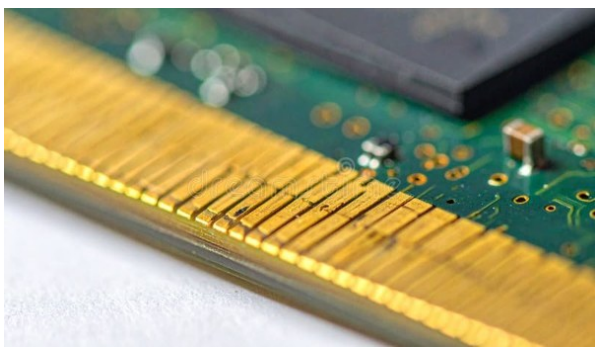


Figure 08 — 01_ram_gold_pins_damage (Échantillon #8)



Figure 09 — 01_ram_gold_pins_damage (Échantillon #9)

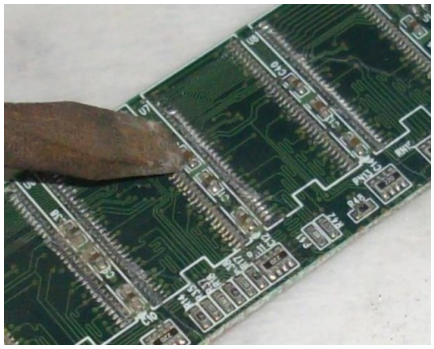


Figure 10 — 01_ram_gold_pins_damage (Échantillon #10)